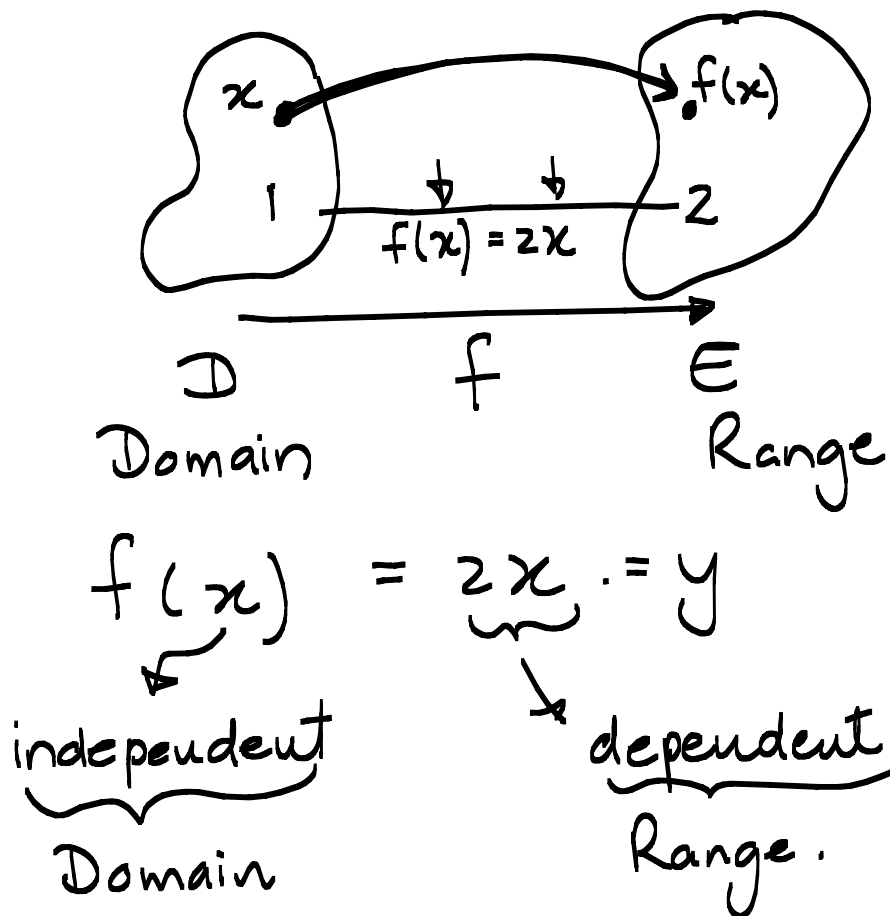


What is a function? p 10.

A function f is a rule that assigns to each element x in a set D exactly one element, $f(x)$ in a set E .

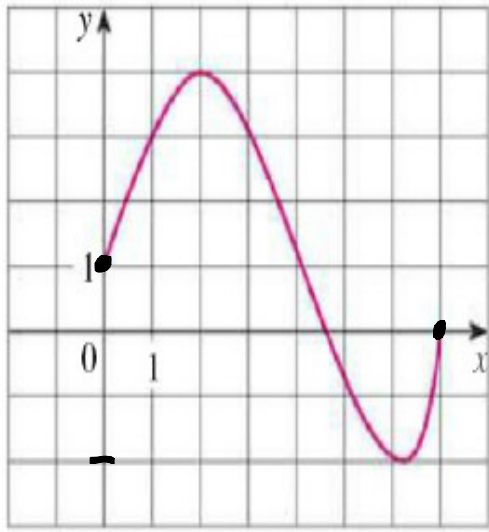


Domain — elements that I may use.

Range — all the elements that I end up with.

Example 1 p 11

①



$$f(1) = 3$$

$$f(5) = -0.7$$

$$\mathcal{D}_f = [0, 7]$$

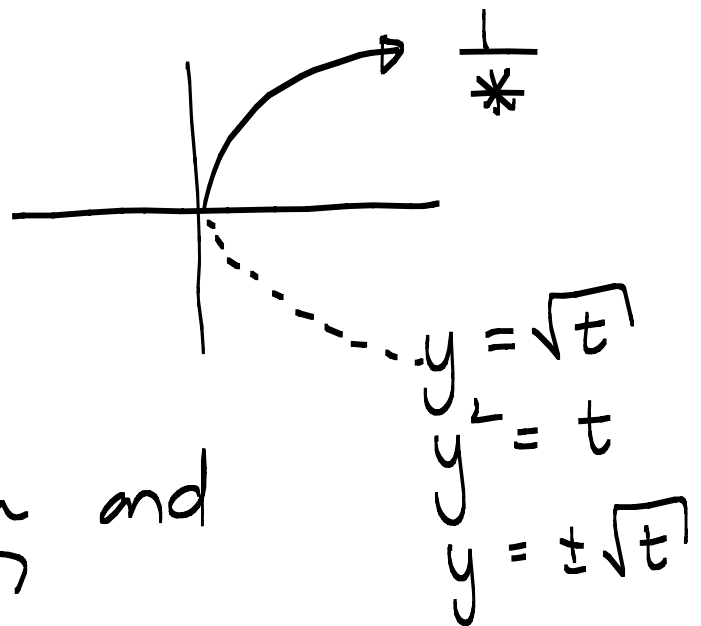
$$\mathcal{R}_f = [-2, 4]$$

② Determine the domain and range of $f(t) = \sqrt{t}$

$$t \geq 0 \quad \text{so}$$

$$\mathcal{D}_f = [0, \infty)$$

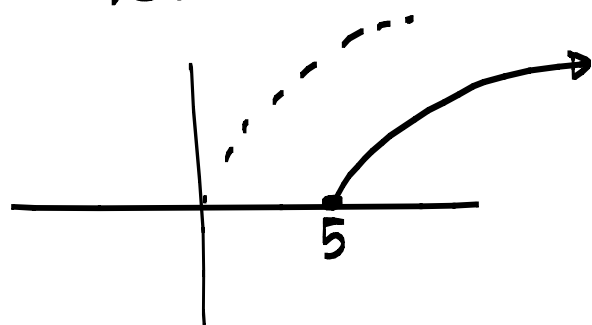
$$\mathcal{R}_f = [0, \infty)$$



③ Find the domain and sketch $g(x) = \sqrt{x-5}$

$$x-5 \geq 0 \quad \text{so} \quad x \geq 5$$

$$\mathcal{D}_g = [5, \infty)$$



④ Determine the domain & range of

$$g(x) = \frac{1}{x^2 - x}$$

$$\text{So } x^2 - x \neq 0 \Rightarrow x(x-1) \neq 0$$

$$\Rightarrow x \neq 0, \text{ and } x \neq 1$$

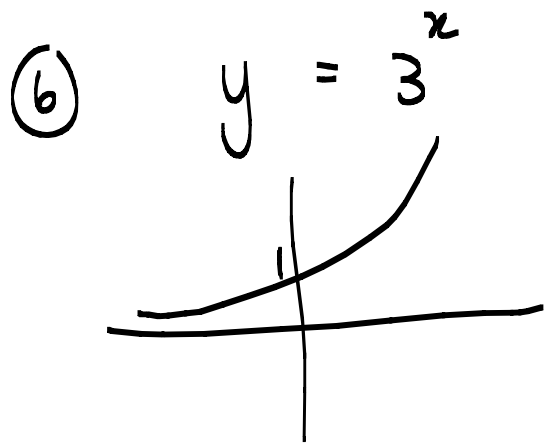
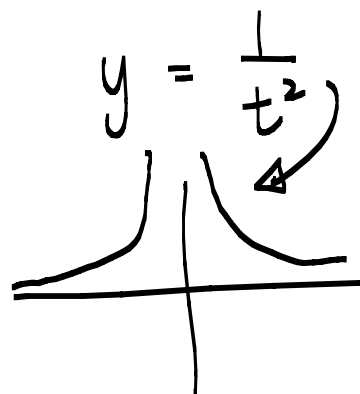
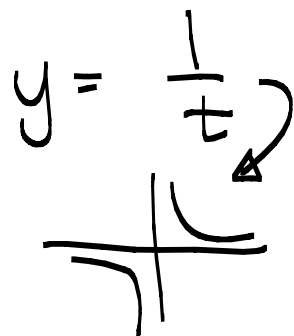
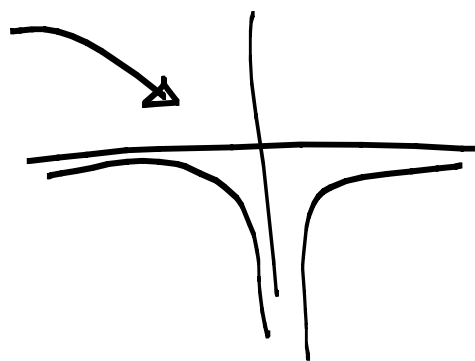
$$D_g = (-\infty, 0) \cup (0, 1) \cup (1, \infty) \text{ or } D_g = \mathbb{R} \setminus \{0, 1\}$$

$$R_g = \mathbb{R} \setminus \{0\} = (-\infty, 0) \cup (0, \infty)$$

$$\textcircled{5} \quad f(t) = -\frac{1}{t^2}$$

$$D_f = \mathbb{R} \setminus \{0\}$$

$$R_f = (-\infty, 0)$$



$$D = \mathbb{R}$$

$$R = (0, \infty)$$

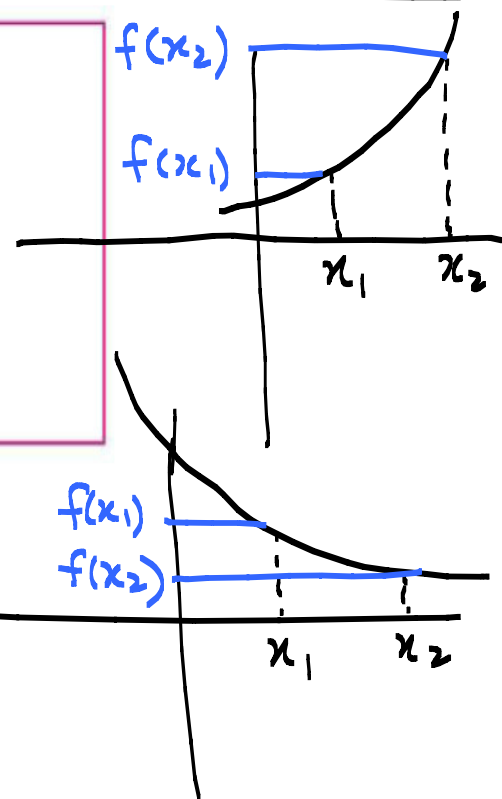
Increasing & Decreasing functions p. 19

A function f is called **increasing** on an interval I if

$$f(x_1) < f(x_2) \quad \text{whenever } x_1 < x_2 \text{ in } I$$

It is called **decreasing** on I if

$$f(x_1) > f(x_2) \quad \text{whenever } x_1 < x_2 \text{ in } I$$

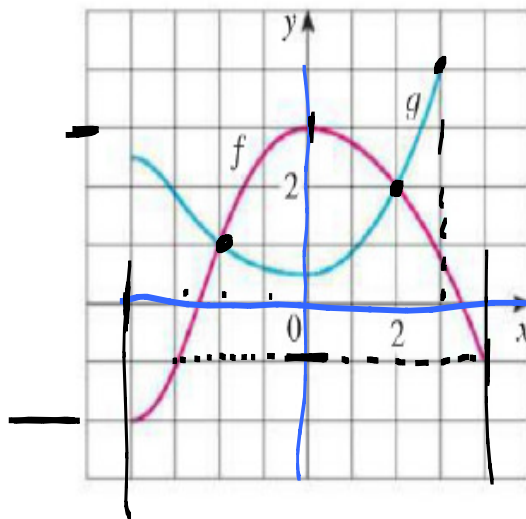


P19

4. The graphs of f and g are given.

- (a) State the values of $f(-4)$ and $g(3)$.
- (b) For what values of x is $f(x) = g(x)$?

- (c) Estimate the solution of the equation $f(x) = -1$.
- (d) On what interval is f decreasing?
- (e) State the domain and range of f .
- (f) State the domain and range of g .



Representation of a function. p12

A. Words

Area of a circle depends on the length of the radius.

$$A = \pi r^2$$
$$f(r) = A$$

B. Numerically (Table)

r	1	0	2
A	π	0	4π

C. Visually (Graph)



D. Algebraically

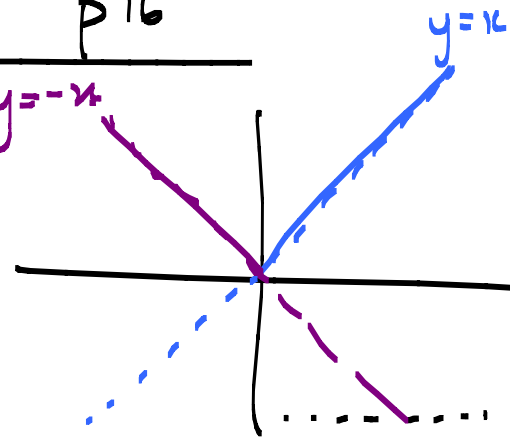
$$* \quad A = \pi r^2 \quad A = f(r) \quad *$$

Vertical line test p15

Unit 1.3 Continued:

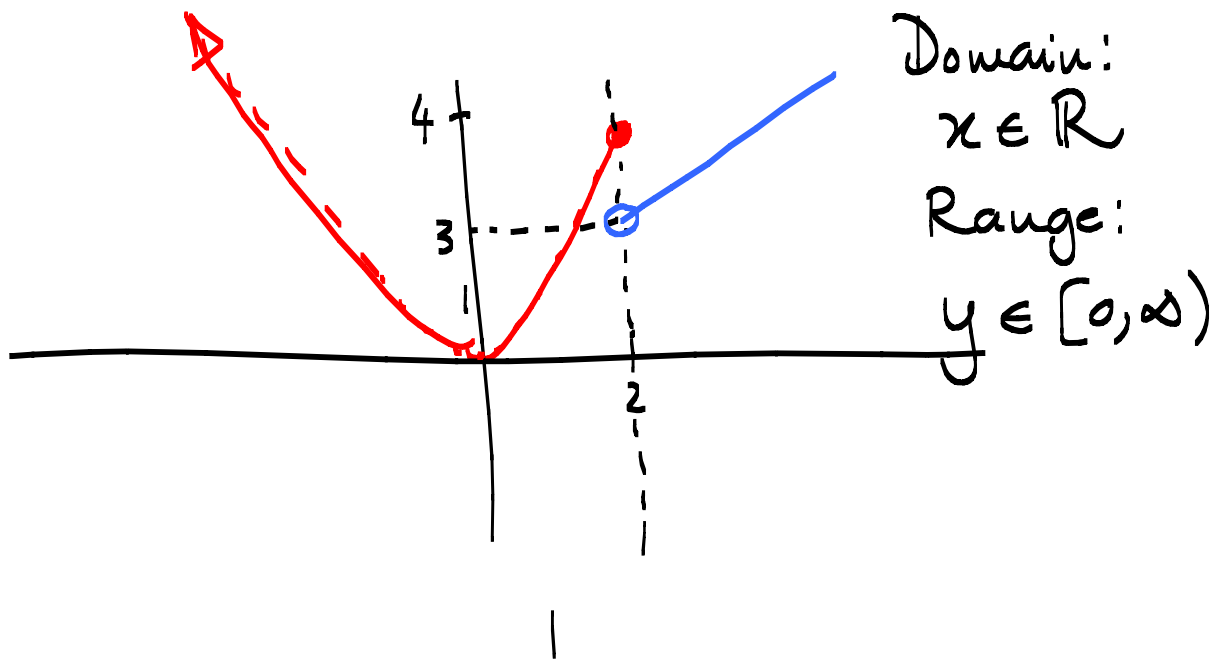
Piecewise defined functions p16

1. $y = |x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}$



2. Sketch the function and determine the domain and range:

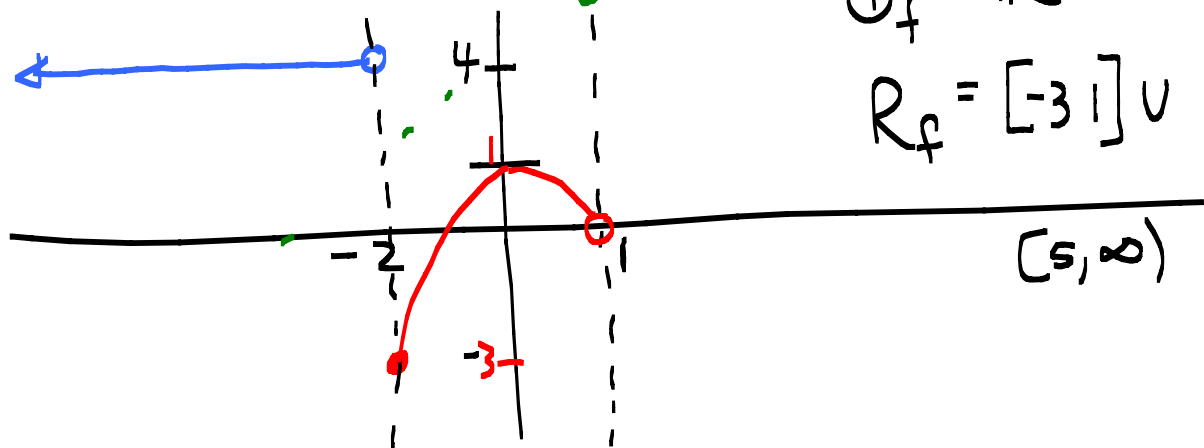
(i) $y = \begin{cases} 1+x & \text{if } x > 2 \\ x^2 & \text{if } x \leq 2 \end{cases}$



$$(ii) \quad f(x) = \begin{cases} 4 & \text{if } x < -2 \\ -x^2 + 1 & \text{if } -2 \leq x < 1 \\ x + 4 & \text{if } x \geq 1 \end{cases}$$

$$\mathcal{D}_f = \mathbb{R}$$

$$R_f = [-3, 1] \cup \{4\} \cup$$

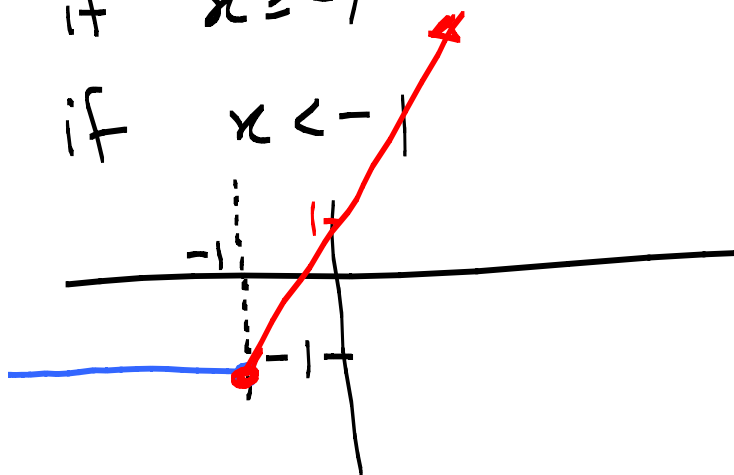


$$(iii) \quad f(x) = x + \underbrace{|x+1|}$$

$$|\underline{x+1}| = \begin{cases} \underline{x+1} & \text{if } \underline{x+1} \geq 0 \quad (x \geq -1) \\ -(x+1) & \text{if } x+1 < 0 \quad (x < -1) \end{cases}$$

$$f(x) = \begin{cases} x + (x+1) & \text{if } x \geq -1 \\ x + [-(x+1)] & \text{if } x < -1 \end{cases}$$

$$= \begin{cases} 2x + 1 & \text{if } x \geq -1 \\ -1 & \text{if } x < -1 \end{cases}$$

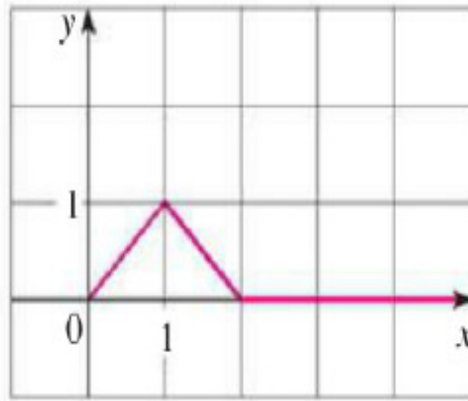


(iv)

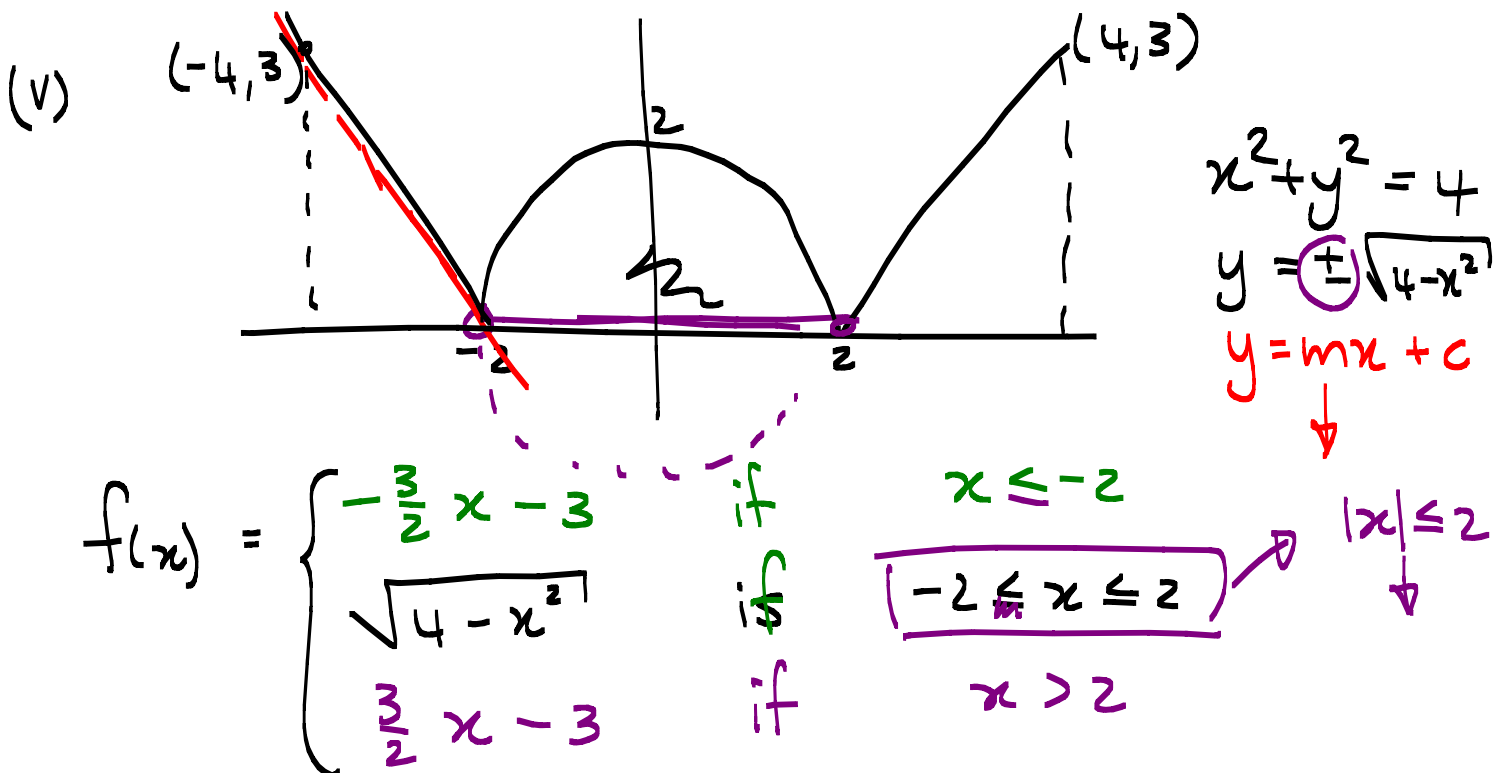
EXAMPLE 9

Find a formula for the function f graphed in Figure 17.

P17



$$f(x) = \begin{cases} x & \text{if } 0 \leq x < 1 \\ -x + 2 & \text{if } 1 \leq x < 2 \\ 0 & \text{if } x \geq 2 \end{cases}$$



Symmetry

Even function:

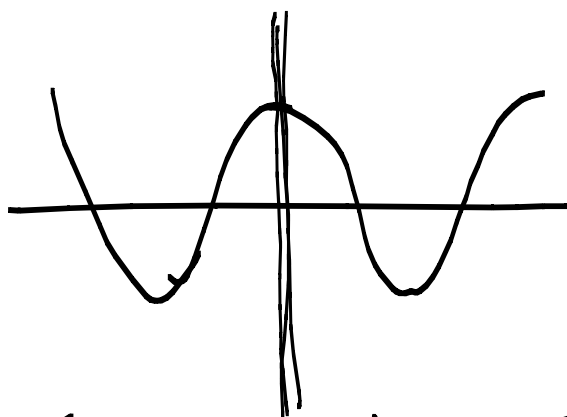
If $f(-x) = f(x) \quad \forall x$ in the domain of f .

$$f(x) = \cos x$$

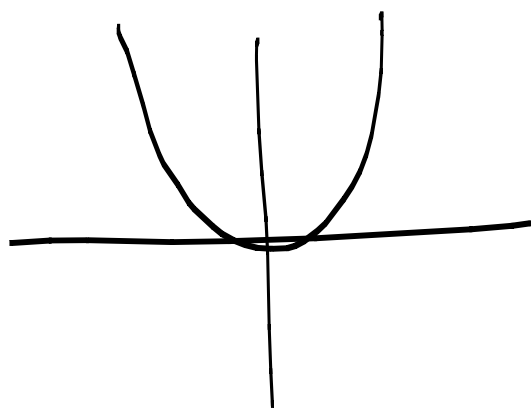
$$f(-x) = \cos(-x) = \cos x$$

$$f(x) = x^2$$

$$f(-x) = (-x)^2 = x^2$$



Symmetrical about y-axis



Odd function:

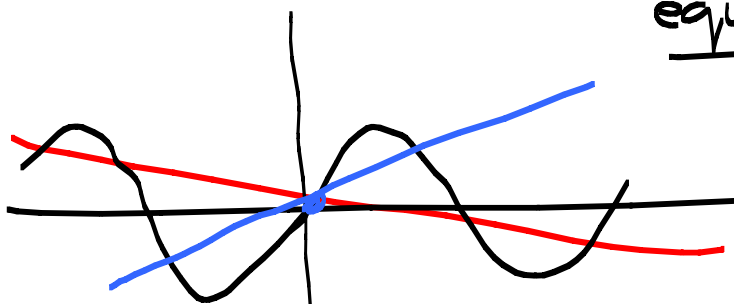
If $f(-x) = -f(x) \quad \forall x$ in the domain of f .

$$f(x) = \sin x$$

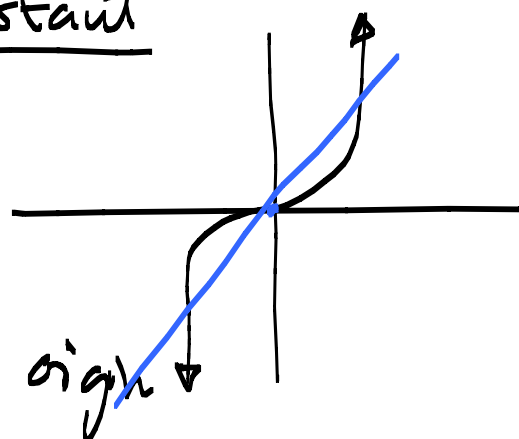
$$f(-x) = \sin(-x) = -\sin x$$

$$f(x) = x^3$$

$$f(-x) = (-x)^3 = -x^3$$



equidistant



⊗ Symmetrical about the origin

Odd or even or neither?

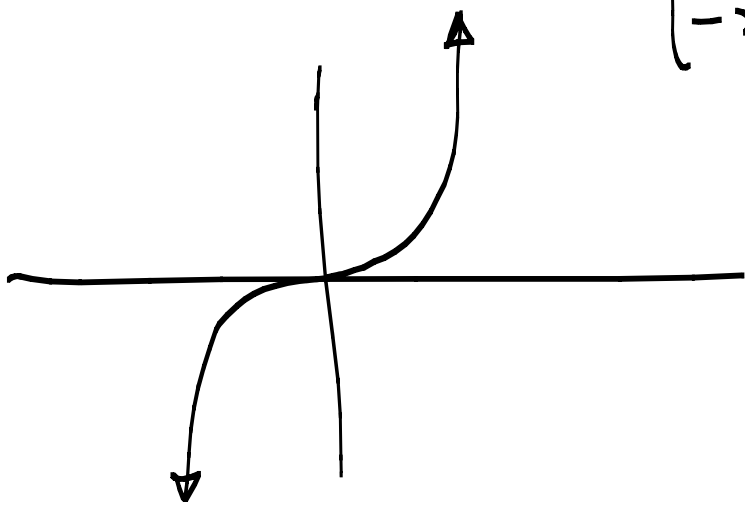
$$(a) \quad h(x) = 2x - x^2 \quad -h(x) = -2x + x^2$$

$$\text{Consider } h(-x) = 2(-x) - (-x)^2 = -2x - x^2$$

Not odd nor even

(b) Is $x|x|$ odd/even?

$$f(x) = x|x| = \begin{cases} x(x) & \text{if } x \geq 0 \\ x(-x) & \text{if } x < 0 \end{cases}$$
$$= \begin{cases} x^2 & \text{if } x \geq 0 \\ -x^2 & \text{if } x < 0 \end{cases}$$



looks like
 f is odd.

Since $f(x) = x|x|$ we have

$$\begin{aligned} \wedge f(-x) &= -x|-x| \\ &= -x|x| \\ &= -f(x) \end{aligned}$$

$\Rightarrow f$ is odd.